

$$=6$$

Assignment II

Paul Haskins
22338905

March 24, 2026

Contents

Chapter	Question I PG 74 notes	Page 1
Chapter	Question II	Page 4
2.1	Part (a)	4
Chapter	Question III pg 87 notes	Page 7
Chapter	Question IV	Page 9

Note I will attach a combined pdf of the lecture notes and I cite some results from it

Chapter 1

Question I PG 74 notes

Radiated power in the leading (quadrupole) approximation is given by

$$P = \frac{G}{45} \frac{d^3 Q_{ij}}{dt^3} \frac{d^3 Q^{ij}}{dt^3} \quad (1.1)$$

and we have that

$$Q^{ij} = q^{ij} - \frac{1}{3} \delta^{ij} q^k_k$$
$$q^{ij} = 3 \int d^3 y \, y^i y^j T_{00}(t, \vec{y}).$$

A star of mass M is orbiting a much heavier star of mass M_H . Compute the power radiated by gravitational waves in the leading (quadrupole) approximation. Assume that the heavy star is stationary, the radius of the orbit is R , and the star motion is described by non-relativistic Newtonian physics.

We have that for a point mass $T_{00}(t, \vec{y}) = M \delta^{(3)}(\vec{y} - \vec{x}(t))$

Thus

$$q^{ij} = 3 \int d^3 y \, y^i y^j T_{00}(t, \vec{y})$$
$$q^{ij} = 3 \int d^3 y \, y^i y^j M \delta^{(3)}(\vec{y} - \vec{x}(t))$$
$$q^{ij} = 3 x^i(t) x^j(t) M.$$

So we have

$$Q^{ij} = 3 x^i(t) x^j(t) M - \frac{1}{3} \delta^{ij} [3 x^k(t) x_k(t) M]$$
$$Q^{ij} = M [3 x^i x^j - \delta^{(i,j)} x^k x_k]$$
$$\dot{Q}^{(i,j)} = M [3 (\dot{x}^i x^j + x^i \dot{x}^j) - \delta^{(i,j)} (2 x^k \dot{x}_k)]$$
$$\ddot{Q}^{(i,j)} = M [3 (\ddot{x}^i x^j + 2 \dot{x}^i \dot{x}^j + x^i \ddot{x}^j) - \delta^{(i,j)} (2 (\dot{x}^k \dot{x}_k + x^k \ddot{x}_k))]$$
$$\ddot{Q}^{(i,j)} = M [3 (\ddot{x}^i x^j + \ddot{x}^i x^j + 2 (\dot{x}^i \dot{x}^j + x^i \ddot{x}^j) + \dot{x}^i \ddot{x}^j + x^i \ddot{x}^j) - \delta^{(i,j)} (2 (\dot{x}^k \dot{x}_k + 2 \dot{x}^k \ddot{x}_k + x^k \ddot{x}_k))].$$

Now lets look at the scalar final term

$$\ddot{x}^k \dot{x}_k + 2 \dot{x}^k \ddot{x}_k + x^k \ddot{x}_k.$$

We can compute the contraction given ($C := \cos(\omega t)$, $S := \sin(\omega t)$ and $\omega = \sqrt{\frac{GM_H}{R^3}}$)

$$x^\ell = \begin{bmatrix} RC \\ RS \\ 0 \end{bmatrix} \quad \dot{x}^\ell = \begin{bmatrix} -R\omega S \\ R\omega C \\ 0 \end{bmatrix} \quad \ddot{x}^\ell = \begin{bmatrix} -R\omega^2 C \\ -R\omega^2 S \\ 0 \end{bmatrix} \quad \ddot{\ddot{x}}^\ell = \begin{bmatrix} R\omega^3 S \\ -R\omega^3 C \\ 0 \end{bmatrix}.$$

We have

$$\begin{aligned} & \ddot{x}^k \dot{x}_k + 2\dot{x}^k \ddot{x}_k + x^k \ddot{\ddot{x}}_k \\ &= [R^2 \omega^3 SC - R^2 \omega^3 CS] + 2[R^2 \omega^3 SC - R^2 \omega^3 CS] + [R^2 \omega^3 CS - R^2 \omega^3 SC] = 0. \end{aligned}$$

So we have the simplification

$$\ddot{Q}^{(i,j)} = M \left[3 \left(\ddot{x}^i x^j + \ddot{x}^i \dot{x}^j + 2 \left(\dot{x}^i \dot{x}^j + \dot{x}^i \ddot{x}^j \right) + \dot{x}^i \ddot{x}^j + x^i \ddot{\ddot{x}}^j \right) \right]$$

and

$$\ddot{Q}_{(i,j)} = M \left[3 \left(\ddot{x}_i x_j + \ddot{x}_i \dot{x}_j + 2 \left(\dot{x}_i \dot{x}_j + \dot{x}_i \ddot{x}_j \right) + \dot{x}_i \ddot{x}_j + x_i \ddot{\ddot{x}}_j \right) \right]$$

$$P = \frac{G}{45} \frac{d^3 Q_{ij}}{dt^3} \frac{d^3 Q^{ij}}{dt^3} \quad (1.2)$$

$$\begin{aligned} \frac{45P}{G} &= \ddot{Q}_{(i,j)} \ddot{Q}^{(i,j)} \\ \frac{5P}{GM^2} &= \left(\ddot{x}_i x_j + \ddot{x}_i \dot{x}_j + 2 \left(\dot{x}_i \dot{x}_j + \dot{x}_i \ddot{x}_j \right) + \dot{x}_i \ddot{x}_j + x_i \ddot{\ddot{x}}_j \right)^2. \end{aligned}$$

If we square out the right hand side where we can drop incidices since all of them are contracted

$$\begin{aligned} RHS &= (\ddot{x} \cdot \ddot{x})(x \cdot x) + 3(\ddot{x} \cdot \ddot{x})(x \cdot \dot{x}) + 3(\ddot{x} \cdot \dot{x})(x \cdot \ddot{x}) + (\ddot{x} \cdot x)(x \cdot \ddot{x}) \\ &+ 3(\ddot{x} \cdot \ddot{x})(\dot{x} \cdot x) + 9(\ddot{x} \cdot \ddot{x})(\dot{x} \cdot \dot{x}) + 9(\ddot{x} \cdot \dot{x})(\dot{x} \cdot \ddot{x}) + 3(\ddot{x} \cdot x)(\dot{x} \cdot \ddot{x}) \\ &+ 3(\dot{x} \cdot \ddot{x})(\ddot{x} \cdot x) + 9(\dot{x} \cdot \ddot{x})(\ddot{x} \cdot \dot{x}) + 9(\dot{x} \cdot \dot{x})(\ddot{x} \cdot \ddot{x}) + 3(\dot{x} \cdot x)(\ddot{x} \cdot \ddot{x}) \\ &+ (x \cdot \ddot{x})(\ddot{x} \cdot x) + 3(x \cdot \ddot{x})(\ddot{x} \cdot \dot{x}) + 3(x \cdot \dot{x})(\ddot{x} \cdot \ddot{x}) + (x \cdot x)(\ddot{x} \cdot \ddot{x}). \end{aligned}$$

But since we are in circular motion we know $x \cdot \dot{x} = 0$ and $x \cdot \ddot{x} = 0$.

The only other combinations are

$$\begin{aligned} (\ddot{x} \cdot \ddot{x}) &= R^2 \omega^6 & \ddot{x} \cdot \ddot{x} &= R^2 \omega^4 \\ (x \cdot x) &= R^2 & \dot{x} \cdot \dot{x} &= R^2 \omega^2 \\ x \cdot \ddot{x} &= -R^2 \omega^2 & \ddot{x} \cdot \dot{x} &= -R^2 \omega^4 \end{aligned}$$

We are left with

$$\begin{aligned} RHS &= R^2 \omega^6 \cdot R^2 + 3(-R^2 \omega^4)(-R^2 \omega^2) + 9(R^2 \omega^4)(R^2 \omega^2) \\ &+ 3(-R^2 \omega^2)(-R^2 \omega^4) + 3(-R^2 \omega^4)(-R^2 \omega^2) + 9(R^2 \omega^2)(R^2 \omega^4) \\ &+ 3(-R^2 \omega^2)(-R^2 \omega^4) + R^2 \cdot R^2 \omega^6 \\ RHS &= 32R^4 \omega^6. \end{aligned}$$

So we have

$$P = \frac{GM^2}{5} (32R^4\omega^6)$$
$$P = \frac{32G^4M^2M_H^3}{5R^5}.$$

Chapter 2

Question II

Given $\Lambda < 0$, $t = x^0$

$$ds^2 = -e^{2\alpha(r)} dt^2 + e^{2\beta(r)} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \quad (2.1)$$

Where the only non-vanishing components of the Ricci tensor is

$$\begin{aligned} R_{00} &= e^{2(\alpha-\beta)} \left(\alpha'' + (\alpha')^2 - \alpha' \beta' + \frac{2}{r} \alpha' \right) \\ R_{rr} &= -\alpha'' - (\alpha')^2 + \alpha' \beta' + \frac{2}{r} \beta' \\ R_{\theta\theta} &= e^{-2\beta} (r (\beta' - \alpha') - 1) + 1 \\ R_{\phi\phi} &= \sin^2 \theta R_{\theta\theta} = \sin^2 \theta (e^{-2\beta} (r (\beta' - \alpha') - 1) + 1). \end{aligned}$$

where $f' := \frac{df}{dr}$ and $\alpha = \alpha(r)$, $\beta = \beta(r)$.

2.1 Part (a)

$$S(g_{\mu\nu}) = \frac{1}{16\pi G_N} \int d^4x \sqrt{-g} (R - 2\Lambda)$$

Equations of motion follow from $\delta S = 0$

$$\begin{aligned} \delta S(g_{\mu\nu}) &= \frac{1}{16\pi G_N} \int d^4x \left[\delta(\sqrt{-g}) (R - 2\Lambda) + \sqrt{-g} (\delta R) \right] \\ \delta S(g_{\mu\nu}) &= \frac{1}{16\pi G_N} \int d^4x \left[\delta\sqrt{-g} (R - 2\Lambda) + \sqrt{-g} (\delta(g^{\mu\nu} R_{\mu\nu})) \right] \end{aligned}$$

We recall that

$$\begin{aligned} \sqrt{-g} &= \sqrt{-e^{Tr(\ln(g_{\mu\nu}))}} = \sqrt{-e^{-Tr(\ln(g^{\mu\nu}))}} \\ \delta\sqrt{-g} &= \delta \left(-e^{-\frac{1}{2} Tr(\ln(g^{\mu\nu}))} \right) = \left(-e^{-\frac{1}{2} Tr(\ln(g^{\mu\nu}))} \right) \cdot \delta \left(-\frac{1}{2} Tr(\ln(g^{\mu\nu})) \right) \\ &= -\frac{1}{2} \sqrt{-g} \delta (Tr(\ln g^{\mu\nu})). \end{aligned}$$

So

$$\delta S(g_{\mu\nu}) = \frac{1}{16\pi G_N} \int d^4x \left[\left(-\frac{1}{2} \sqrt{-g} \delta(\text{Tr}(\ln g^{\mu\nu})) \right) (R - 2\Lambda) + \sqrt{-g} R_{\mu\nu} (\delta g^{\mu\nu}) + \sqrt{-g} g^{\mu\nu} \delta R_{\mu\nu} \right]$$

We can use $\delta \ln A = A^{-1} \delta A$ and $\delta \text{Tr}(A) = \text{Tr}(\delta A)$

$$\delta \sqrt{-g} = -\frac{1}{2} \sqrt{-g} \text{Tr}(g_{\mu\nu} \delta g^{\mu\nu}) \quad (2.2)$$

We also note that we were given in HW 1 Q4 " $\delta R = R_{\mu\nu} \delta g^{\mu\nu}$ up to a total derivative". Thus we drop the $g_{\mu\nu} \delta R^{\mu\nu}$ term. Combining these we get

$$\begin{aligned} \delta S(g_{\mu\nu}) &= \frac{1}{16\pi G_N} \int d^4x \left[\left(-\frac{1}{2} \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu} \right) (R - 2\Lambda) + \sqrt{-g} R_{\mu\nu} (\delta g^{\mu\nu}) \right] \\ \delta S(g_{\mu\nu}) &= \frac{1}{16\pi G_N} \int d^4x \left[\left(-\frac{1}{2} \sqrt{-g} g_{\mu\nu} \right) (R - 2\Lambda) + \sqrt{-g} R_{\mu\nu} \right] \delta g^{\mu\nu} \end{aligned}$$

The terms in the parenthesis must vanish, thus we can drop the prefactor.

$$0 = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} (R - 2\Lambda).$$

If we contract with $g^{\mu\nu}$ we obtain

$$R = \frac{1}{2} (4) (R - 2\Lambda) \implies R = 4\Lambda \quad (2.3)$$

Thus we have

$$R_{\mu\nu} = g_{\mu\nu} \Lambda.$$

So we can see that $\kappa = 1$.

One can take any two components of $R_{\mu\nu} = g_{\mu\nu} \Lambda$ and divide by the (scalar) component of $g_{\mu\nu}$.

That quotient must equal Λ . For example I will take $\frac{R_{00}}{g_{00}} = \Lambda$ and $\frac{R_{rr}}{g_{rr}} = \Lambda$.

If we write the explicitly we have

$$\begin{aligned} \frac{R_{00}}{g_{00}} &= e^{-2\alpha} \left[e^{2(\alpha-\beta)} \left(\alpha'' + (\alpha')^2 - \alpha' \beta' + \frac{2}{r} \alpha' \right) \right] = -\Lambda \\ \frac{R_{rr}}{g_{rr}} &= e^{-2\beta} \left[-\alpha'' - (\alpha')^2 + \alpha' \beta' + \frac{2}{r} \beta' \right] = \Lambda. \end{aligned}$$

Adding these equations gives us

$$\begin{aligned} \alpha' + \beta' &= 0 \\ \alpha &= -\beta + c. \end{aligned}$$

To see why we can set $c = 0$ one can consider the transformation $t \rightarrow \tilde{t} := e^c t$ thus $dt^2 \rightarrow e^{-2c} d\tilde{t}^2$. We can then identify $g_{00} = -e^{2(\alpha-c)}$. One can define $\tilde{\alpha} = \alpha - c$. Then we notice $\tilde{\alpha} = -\beta$. Nothing else in the metric depends on t so c is unphysical. Therefore we can drop the tildes and we have $\alpha = -\beta$ WLOG.

If we have $\alpha = -\beta$ we can define the function $f(r) = e^{2\alpha} = e^{-2\beta}$ and its derivative $f'(r) = 2\alpha'e^{2\alpha} = -2\beta'e^{-2\beta}$

Now we can look at $R_{\theta\theta} = \Lambda g_{\theta\theta}$, Explicitly

$$\begin{aligned} R_{\theta\theta} &= e^{-2\beta} (r(\beta' - \alpha') - 1) + 1 = r^2\Lambda \\ r^2\Lambda &= f(-r(\alpha' - \beta') - 1) + 1 \\ r^2\Lambda &= -rf' - f + 1. \end{aligned}$$

Now we can write both sides as a total derivative and integrate

$$\begin{aligned} d\left(\frac{1}{3}r^3\Lambda\right) &= d(-rf + r) \\ \frac{1}{3}r^3\Lambda &= -rf + r + c \\ -rf &= \frac{1}{3}r^3\Lambda - r - c \\ f &= -\frac{1}{3}r^2\Lambda + 1 + \frac{c}{r}. \end{aligned}$$

If we consider the weak field limit (Λ is negligible and r is large, but not too large that the r^2 would dominate) we know the Schwarzschild solution takes the well known form $g_{00} = -\left(1 - \frac{2GM}{r}\right)$ due to the fact that $g_{00} \approx -(1 + 2\Phi)$ where $\Phi = \frac{-GM}{r}$. This matches Newton in the equation for acceleration due to gravitational attraction $a = -\nabla\Phi$. We can make the identification $c \leftrightarrow -2GM$.

So we have

$$e^{2\alpha} = 1 - \frac{2GM}{r} - \frac{1}{3}r^2\Lambda.$$

and $\Lambda < 0$, so we have the Anti De-Sitter Schwarzschild solution.

Chapter 3

Question III pg 87 notes

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) dt^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \quad (3.1)$$

As we saw in the lecture notes we can use the geodesics

$$\frac{dx^\mu}{d\lambda^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{d\lambda} \frac{dx^\beta}{d\lambda} = 0 \quad (3.2)$$

We define the killing vector $\mathcal{K}^\mu := (1, 0, 0, 0)$ which gives us the conservation of a quantity (energy here).

This gives us

$$E = -g_{\mu\nu} \mathcal{K}^\mu u^\nu = \left(1 - \frac{2GM}{r}\right) \frac{dt}{d\lambda} \quad \text{where } \frac{dt}{d\lambda} = u^0 \quad (3.3)$$

At asymptotic infinity the metric is flat. Therefore, the 4 velocity of the astronaut is $u^\mu = (\gamma_0, -\gamma_0 u_0, 0, 0)$, $\gamma_0 := \frac{1}{\sqrt{1-u_0^2}}$. We have $-d\lambda^2 = g_{\mu\nu} dx^\mu dx^\nu$ so $-1 = g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda}$. Thus we have

$$-1 = g_{\mu\nu} s^\mu s^\nu$$

If we find the energy at infinity

$$E = \gamma_0 = \frac{1}{\sqrt{1-u_0^2}} \quad (3.4)$$

If we denote the stationary observers 4 velocity by s^μ we can identify that since they are stationary $s^i = 0$ where $i = 1, 2, 3$ while $s^0 \neq 0$.

$$-1 = g_{00} s^0 s^0 \implies s_0 = -\left(1 - \frac{2GM}{r}\right)^{-\frac{1}{2}} \quad (3.5)$$

The speed the observer measures comes from the local lorentz factor

$$\gamma = -g_{\mu\nu} u^\mu s^\nu \quad (3.6)$$

$$\begin{aligned} \gamma &= -\left(1 - \frac{2GM}{r}\right) \frac{E}{1 - \frac{2GM}{r}} \left(1 - \frac{2GM}{r}\right)^{-\frac{1}{2}} (-1) \\ \gamma &= \frac{E}{\sqrt{1 - \frac{2GM}{r}}} = \frac{\gamma_0}{\sqrt{1 - \frac{2GM}{r}}} \quad \text{so } \gamma^2 = \frac{\gamma_0^2}{1 - \frac{2GM}{r}} = \frac{1}{\left(1 - \frac{2GM}{r}\right)} \frac{1}{1 - u_0^2}. \end{aligned}$$

Now we can solve $v(\gamma)$ and substitute this expression for γ .

$$\begin{aligned}\gamma &= \frac{1}{\sqrt{1-v^2}} \quad \Rightarrow \quad v = \sqrt{1 - \frac{1}{\gamma^2}} \\ v &= \sqrt{1 - \left(1 - \frac{2GM}{r}\right) (1 - u_0^2)} \\ v &= \sqrt{u_0^2 + \frac{2GM}{r} (1 - u_0^2)}.\end{aligned}$$

We can notice if we test initial velocities we are consistent with physical requirements. For example if $u_0 = 0$ we have $v = \sqrt{\frac{2GM}{r}}$ which is the newtonian escape velocity. If we have $u_0 = 1$ (a photon) we have $v = 1$ the speed is a constant (namely the speed of light).

Chapter 4

Question IV

Newtonian physics states that the period of a satellite is given by

$$T^2 = \frac{4\pi^2 r^3}{GM} \quad (4.1)$$

we have $r = 7GM$

$$T = \sqrt{\frac{4\pi^2 (7GM)^3}{GM}} = \sqrt{7} (14\pi GM) \quad (4.2)$$

We would expect for the observer at infinity to match this.

The Schwarzschild lagrangian

$$\mathcal{L} = \frac{1}{2} g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu \quad (4.3)$$

$$\mathcal{L} = \frac{1}{2} \left[- \left(1 - \frac{2GM}{r} \right) \dot{t}^2 + \left(1 - \frac{2GM}{r} \right)^{-1} \dot{r}^2 + r^2 \dot{\phi}^2 \right] \quad (4.4)$$

We have t and ϕ cyclic coords

The euler lagrange eqns give us

$$\left(1 - \frac{2GM}{r} \right) \frac{dt}{d\tau} = E \quad r^2 \frac{d\phi}{d\tau} = L \quad (4.5)$$

where E and L are constants (Energy and Angular momentum).

$$ds^2 = - \left(1 - \frac{2GM}{r} \right) dt^2 + \left(1 - \frac{2GM}{r} \right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \quad (4.6)$$

For a massive particle $ds^2 = -d\tau^2$, so substituting and dividing by $d\tau^2$ in 4.6 gives

$$-1 = - \left(1 - \frac{2GM}{r} \right) \left(\frac{dt}{d\tau} \right)^2 + \left(1 - \frac{2GM}{r} \right)^{-1} \left(\frac{dr}{d\tau} \right)^2 + r^2 \left(\frac{d\phi}{d\tau} \right)^2 \quad (4.7)$$

We can use 4.5

$$-1 = - \left(1 - \frac{2GM}{r} \right) \left[E \left(1 - \frac{2GM}{r} \right)^{-1} \right]^2 + \left(1 - \frac{2GM}{r} \right)^{-1} \left(\frac{dr}{d\tau} \right)^2 + r^2 \left(\frac{L}{r^2} \right)^2 \quad (4.8)$$

Thus

$$\left(\frac{dr}{dt}\right)^2 = -\left(1 - \frac{2GM}{r}\right)\left(1 + \frac{L^2}{r^2}\right) + E^2 \quad (4.9)$$

We can identify the effective potential as in ACM

$$V_{\text{eff}} = \left(1 - \frac{2GM}{r}\right)\left(1 + \frac{L^2}{r^2}\right) \quad (4.10)$$

Since we have a circular orbit we know that

$$\frac{dr}{dt} = 0 \quad \frac{dV_{\text{eff}}}{dr} = 0 \quad (4.11)$$

We have

$$\begin{aligned} \frac{dV}{dr} &= \left(\frac{2GM}{r^2}\right)\left(1 + \frac{L^2}{r^2}\right) - 2\left(1 - \frac{2GM}{r}\right)\left(\frac{L^2}{r^3}\right) = 0 \\ 0 &= 2GMr\left(1 + \frac{L^2}{r^2}\right) - 2\left(1 - \frac{2GM}{r}\right)L^2 \\ 0 &= 2GMr + \frac{2GM}{r}L^2 - 2L^2 + \frac{4GM}{r}L^2 \\ -2GMr^2 &= (6GM - 2r)L^2 \end{aligned}$$

We get

$$L^2 = \frac{GMr^2}{r - 3GM} \quad (4.12)$$

And we know $E^2 = V_{\text{eff}}$

$$\begin{aligned} V_{\text{eff}} &= \left(1 - \frac{2GM}{r}\right)\left(1 + \frac{L^2}{r^2}\right) \\ &= \left(1 - \frac{2GM}{r}\right)\left(1 + \frac{1}{r^2}\left(\frac{GMr^2}{r - 3GM}\right)\right) \\ &= \left(1 - \frac{2GM}{r}\right)\left(1 + \frac{GM}{r - 3GM}\right) \\ &= 1 + \frac{GM}{r - 3GM} - \frac{2G^2M^2}{r(r - 3GM)} - 2\frac{GM}{r} \end{aligned}$$

If we multiply each term such that the denominator of each is $r(r - 3GM)$ we have the numerator N given by

$$\begin{aligned} N &= r(r - 3GM) + rGM - 2G^2M^2 - 2GM(r - 3GM) \\ N &= r^2 - 3GMr + rGM - 2G^2M^2 - 2GMr + 6G^2M^2 \\ N &= r^2 - 4GM + 4G^2M^2 = (r^2 - 2GM)^2. \end{aligned}$$

Reintroducing the denominator gives

$$E^2 = \frac{(r - 2GM)^2}{r(r - 3GM)} \quad (4.13)$$

And we have

$$\frac{d\Omega}{dt} = \frac{\left(\frac{d\phi}{d\tau}\right)}{\left(\frac{dt}{d\tau}\right)} = \frac{d\phi}{dt} = \frac{\left(\frac{L}{r^2}\right)}{\left(\frac{E}{\left(1-\frac{2GM}{r}\right)}\right)} \quad (4.14)$$

Performing the substitutions (sorry I am lazy to type it out) we have

$$\frac{d\phi}{dt} = \sqrt{\frac{GM}{r^3}} \quad (4.15)$$

and

$$T_\infty = \frac{2\pi}{\left(\frac{d\phi}{dt}\right)} = 2\pi\sqrt{\frac{r^3}{GM}} = 14\sqrt{7}\pi GM \quad (4.16)$$

Which exactly matches the Newtonian answer.

The time as observed by the spaceship:

Since it is in a circular orbit $dr = 0$

$$d\tau^2 = \left(1 - \frac{2GM}{r}\right) dt^2 - r^2 d\phi^2 \quad (4.17)$$

$$\begin{aligned} \frac{d\tau^2}{dt^2} &= \left(1 - \frac{2GM}{r}\right) - r^2 \left(\frac{d\phi^2}{dt^2}\right) = \left(1 - \frac{2GM}{r}\right) - r^2 \left(\frac{GM}{r^3}\right) = 1 - \frac{3GM}{r} \\ \frac{d\tau^2}{dt^2} &= \frac{4}{7}. \end{aligned}$$

$$\begin{aligned} \frac{T_{\text{ship}}}{T_\infty} &= \frac{d\tau^2}{dt^2} \implies T_{\text{ship}} = T_\infty \sqrt{1 - \frac{3GM}{r}} \\ T_{\text{ship}} &= \left(14\sqrt{7}\pi GM\right) \left(\frac{2}{\sqrt{7}}\right) = 28\pi GM. \end{aligned}$$